Experiment 2: Online Hotel Room Booking System

1. Aim

To design a responsive front-end prototype of an online hotel room booking website in which guests can check room availability, book, change and cancel rooms, and the administrator can manage rooms, bookings and static pages.

2. Algorithm

1. User enters arrival date, departure date and guests.
2. For each room type, availability = total rooms - bookings that overlap the chosen dates.
3. If availability > 0 the Book Now button is enabled, otherwise Sold out is shown.
4. User fills the booking form and payment details; a booking is created with status Confirmed.
5. For change or cancel: if (arrival date - today) >= 3 days the buttons are enabled, otherwise they are disabled.
6. Admin may edit quantity and price of room types and edit or cancel any booking at any time.

3. Source Code

exp2_hotel.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Hotel Sunrise - Room Booking</title>
<style>
:root{--c:#8a4b08}
*{box-sizing:border-box}
body{margin:0;font-family:"Segoe UI",Arial,sans-serif;background:#f6f3ee;color:#222}
header{background:var(--c);color:#fff;padding:10px 16px;display:flex;flex-wrap:wrap;gap:10px;align-
items:center;justify-content:space-between}
.brand{font-size:1.25rem;font-weight:600}
nav{display:flex;flex-wrap:wrap;gap:4px}
nav a{color:#fff;opacity:.9;padding:6px 10px;border-radius:6px;font-size:.9rem;cursor:pointer}
nav a:hover{background:rgba(255,255,255,.2)}
main{max-width:1000px;margin:20px auto;padding:0 16px}
h1,h2{color:var(--c);margin-top:0}
.card{background:#fff;border-radius:8px;padding:18px;box-shadow:0 1px 4px rgba(0,0,0,.12);margin-
bottom:16px}
.narrow{max-width:440px;margin:24px auto}
label{display:block;margin:10px 0 4px;font-weight:600;font-size:.9rem}
input,select,textarea{width:100%;padding:8px;border:1px solid #bbb;border-radius:5px;font:inherit}
textarea{min-height:110px}
button{background:var(--c);color:#fff;border:0;padding:8px 14px;border-
radius:5px;cursor:pointer;font:inherit}
button.red{background:#b8423a}button.grey{background:#777}button.s{padding:4px 9px;font-size:.82rem}
button:disabled{background:#bbb;cursor:not-allowed}
.row{display:flex;flex-wrap:wrap;gap:10px;align-items:end}.row>*{flex:1;min-width:130px}.row
button{flex:0 0 auto}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:14px}
table{width:100%;border-collapse:collapse;background:#fff}
th,td{padding:8px 10px;border-bottom:1px solid #e5e5e5;text-align:left;font-size:.9rem}
th{background:#f0e6d8}
.tw{overflow-x:auto}
.ok{color:#1d7a35;font-weight:600}.bad{color:#b8423a;font-weight:600}.muted{color:#666;font-
size:.88rem}
.msg{padding:10px 14px;border-radius:6px;margin-
bottom:12px}.msg.s{background:#def3e2;color:#1d5a2a}.msg.e{background:#fbe0de;color:#8a2822}
.big{font-size:2.2rem}
</style>
</head>
<body>
<header><div class="brand">&#127976; Hotel Sunrise</div><nav id="nav"></nav></header>
<main>
<div id="msg"></div>

<!-- ============ PUBLIC PAGES ============ -->
<section class="page" id="home">
```

```
  <div class="card"><h1>Find &amp; Book Your Room</h1>
    <p class="muted">Check room availability for your dates, then book instantly.</p>
    <div class="row">
      <div><label>Arrival date</label><input type="date" id="s_in"></div>
      <div><label>Departure date</label><input type="date" id="s_out"></div>
      <div><label>Guests</label><select id="s_g"><option>1</option><option
selected>2</option><option>3</option><option>4</option></select></div>
      <button onclick="searchRooms()">Check Availability</button>
    </div>
  </div>
  <div id="results" class="grid"></div>
</section>

<section class="page" id="rooms"><div class="card"><h1>Room Types</h1><div
id="rooms_text"></div></div></section>
<section class="page" id="about"><div class="card"><h1>About Us</h1><div
id="about_text"></div></div></section>
<section class="page" id="contactus"><div class="card"><h1>Contact Us</h1><div
id="contactus_text"></div></div></section>
<section class="page" id="qa"><div class="card"><h1>Customer Service Q&amp;amp;A</h1><div
id="qa_text"></div></div></section>
<section class="page" id="travel"><div class="card"><h1>Local Travel &amp; Shipping Guide</h1><div
id="travel_text"></div></div></section>
<section class="page" id="privacy"><div class="card"><h1>Privacy Policy</h1><div
id="privacy_text"></div></div></section>

<section class="page" id="login"><div class="card narrow"><h1>Login</h1>
  <label>Username</label><input id="l_u"><label>Password</label><input id="l_p" type="password">
  <label>Login as</label><select id="l_r"><option value="user">Guest (User)</option><option
value="admin">Administrator</option></select>
  <p><button onclick="doLogin()">Login</button></p>
  <p class="muted">Demo: any username and password works. New guest? <a href="#"
onclick="show('register');return false">Register</a></p></div></section>

<section class="page" id="register"><div class="card narrow"><h1>Register</h1>
  <label>Full name</label><input id="r_n"><label>Email</label><input id="r_e"
type="email"><label>Phone</label><input id="r_ph">
  <label>Username</label><input id="r_u"><label>Password</label><input id="r_p" type="password">
  <p><button onclick="doRegister()">Register</button></p></div></section>

<!-- ============ USER PAGES ============ -->
<section class="page" id="book"><div class="card narrow"><h1>Booking Form</h1>
  <p id="b_sum"></p>
  <label>Guest name</label><input id="b_n"><label>Phone</label><input id="b_ph">
  <h2 style="margin-top:18px">Payment (demo)</h2>
  <label>Card number</label><input id="b_c" placeholder="1234 5678 9012 3456" maxlength="19">
  <div class="row"><div><label>Expiry</label><input
placeholder="MM/YY"></div><div><label>CVV</label><input type="password" maxlength="3"></div></div>
  <p><button onclick="confirmBooking()">Pay &amp; Confirm Booking</button> <button class="grey"
onclick="show('home')">Cancel</button></p></div></section>

<section class="page" id="mybookings"><h1>My Bookings</h1>
  <p class="muted">Bookings can be changed or cancelled up to 3 days before the arrival date.</p><div
class="tw" id="mb_table"></div></section>

<section class="page" id="change"><div class="card narrow"><h1>Change Booking</h1><p id="c_sum"></p>
  <label>New arrival date</label><input type="date" id="c_in"><label>New departure date</label><input
type="date" id="c_out">
  <p><button onclick="saveChange()">Save Changes</button> <button class="grey"
onclick="show('mybookings')">Back</button></p></div></section>

<section class="page" id="profile"><div class="card narrow"><h1>My Profile</h1>
  <label>Full name</label><input id="p_n"><label>Email</label><input
id="p_e"><label>Phone</label><input id="p_ph">
  <p><button onclick="saveProfile()">Update Profile</button></p></div></section>

<!-- ============ ADMIN PAGES ============ -->
<section class="page" id="a_rooms"><h1>Manage Rooms (quantity &amp; price)</h1><div class="tw"
id="ar_table"></div></section>
<section class="page" id="a_bookings"><h1>All Bookings</h1><div class="tw"
id="ab_table"></div></section>
<section class="page" id="a_editbook"><div class="card narrow"><h1>Edit Booking</h1><p id="e_sum"></p>
  <label>Guest name</label><input id="e_n"><label>Room type</label><select id="e_r"></select>
  <label>Arrival</label><input type="date" id="e_in"><label>Departure</label><input type="date"
id="e_out">
  <p><button onclick="saveAdminEdit()">Save</button> <button class="grey"
onclick="show('a_bookings')">Back</button></p></div></section>
```

```
<section class="page" id="a_static"><div class="card"><h1>Edit Static Pages</h1>
  <label>Page</label><select id="st_sel" onchange="loadStatic()"></select>
  <label>Content (HTML allowed)</label><textarea id="st_txt"></textarea>
  <p><button onclick="saveStatic()">Save Page</button></p></div></section>
</main>

<script>
// ---------------------------------------------- helpers
function $(i){return document.getElementById(i)}
function pad(n){return (n<10?'0':'')+n}
function fmt(d){return d.getFullYear()+'-'+pad(d.getMonth()+1)+'-'+pad(d.getDate())}
function addDays(n){var d=new Date();d.setDate(d.getDate()+n);return fmt(d)}
function days(a,b){var x=a.split('-'),y=b.split('-');return Math.round((Date.UTC(+y[0],+y[1]-1,+y[2])-
Date.UTC(+x[0],+x[1]-1,+x[2]))/864e5)}
function msg(t,ok){$('msg').innerHTML='<div class="msg
'+(ok===false?'e':'s')+'">'+t+'</div>';window.scrollTo(0,0)}
function
show(id){document.querySelectorAll('.page').forEach(function(p){p.hidden=p.id!==id});$('msg').innerHTM
L='';window.scrollTo(0,0);if(window['on_'+id])window['on_'+id]();drawNav()}

// ---------------------------------------------- data (in memory)
var ROOMS=[
 {t:'Standard Room',price:2500,qty:10,icon:'&#128716;',info:'Cosy room with a double bed, TV and Wi-
Fi.'},
 {t:'Deluxe Room',price:4000,qty:6,icon:'&#128715;',info:'Larger room with city view, mini-bar and
work desk.'},
 {t:'Executive Suite',price:7500,qty:3,icon:'&#127976;',info:'Bedroom plus living area, bathtub and
breakfast included.'},
 {t:'Family Room',price:5500,qty:4,icon:'&#128106;',info:'Two double beds, sleeps four. Ideal for
families.'}];
var STATIC={
 rooms:'<p>We offer four room types: Standard, Deluxe, Executive Suite and Family Room. All rooms have
air conditioning, free Wi-Fi and 24-hour room service.</p>',
 about:'<p>Hotel Sunrise is a 3-star hotel in the heart of the city, serving guests since 1998.</p>',
 contactus:'<p>Phone: +91 98765 43210<br>Email: stay@hotelsunrise.example<br>Address: 12 Beach Road,
Chennai</p>',
 qa:'<p><b>Check-in time?</b> 12:00 noon. <b>Check-out time?</b> 11:00 am.<br><b>Can I cancel?</b>
Yes, up to 3 days before arrival.<br><b>Is breakfast included?</b> Only in the Executive Suite.</p>',
 travel:'<p>Airport: 20 km (taxi 40 min). Railway station: 5 km. Local sightseeing and
shipping/courier assistance available at the front desk.</p>',
 privacy:'<p>We store only the details needed for your booking and never share them with third
parties.</p>'};
var users=[{u:'guest',n:'Demo Guest',e:'guest@mail.com',ph:'9999999999'}];
var bookings=[
 {id:1001,room:1,in:addDays(2),out:addDays(4),guests:2,name:'Demo
Guest',by:'guest',total:8000,status:'Confirmed'},
 {id:1002,room:0,in:addDays(10),out:addDays(12),guests:2,name:'Demo
Guest',by:'guest',total:5000,status:'Confirmed'}];
var nextId=1003, me=null, pending=null, editing=null;

// ---------------------------------------------- navigation
var PUB=[['home','Home'],['rooms','Rooms'],['about','About'],['qa','Q&A'],['travel','Travel
Guide'],['contactus','Contact'],['privacy','Privacy']];
function drawNav(){
  var l=PUB.slice();
  if(!me) l=l.concat([['login','Login'],['register','Register']]);
  else if(me.role==='user') l=l.concat([['mybookings','My
Bookings'],['profile','Profile'],['out','Logout']]);
  else l=[['a_rooms','Rooms &amp;amp; Prices'],['a_bookings','Bookings'],['a_static','Static
Pages']].concat(PUB.slice(0,1)).concat([['out','Logout']]);
  $('nav').innerHTML=l.map(function(x){return '<a
onclick="'+(x[0]==='out'?'logout()':"show('"+x[0]+"')")+'">'+x[1]+'</a>'}).join('');
}
function logout(){me=null;show('login');msg('Logged out.')}

// ---------------------------------------------- auth
function doLogin(){
  if(!$('l_u').value.trim()||!$('l_p').value)return msg('Enter username and password.',false);
  me={name:$('l_u').value.trim(),role:$('l_r').value,u:$('l_u').value.trim()};
  var f=users.find(function(x){return x.u===me.u});if(!f){f={u:me.u,n:me.u,e:'',ph:''};users.push(f)}
  show(me.role==='admin'?'a_rooms':'home');msg('Welcome, '+me.name+'!');
}
function doRegister(){
  if(!$('r_n').value.trim()||!$('r_u').value.trim()||!$('r_p').value)return msg('Name, username and
password are required.',false);
  users.push({u:$('r_u').value.trim(),n:$('r_n').value.trim(),e:$('r_e').value,ph:$('r_ph').value});
  show('login');msg('Registration successful. Please log in.');
}
```

```
function on_profile(){var f=users.find(function(x){return
x.u===me.u});$('p_n').value=f.n;$('p_e').value=f.e;$('p_ph').value=f.ph}
function saveProfile(){var f=users.find(function(x){return
x.u===me.u});f.n=$('p_n').value;f.e=$('p_e').value;f.ph=$('p_ph').value;me.name=f.n;msg('Profile
updated.')}

// ---------------------------------------------- static pages
function on_rooms(){$('rooms_text').innerHTML=STATIC.rooms+ROOMS.map(function(r){return '<p>'+r.icon+'
<b>'+r.t+'</b> - Rs. '+r.price+'/night<br>'+r.info+'</p>'}).join('')}
['about','contactus','qa','travel','privacy'].forEach(function(k){window['on_'+k]=function(){$(k+'_tex
t').innerHTML=STATIC[k]}});

// ---------------------------------------------- availability + booking
function booked(ri,a,b){return bookings.filter(function(x){return
x.room===ri&&x.status!=='Cancelled'&&x.in<b&&x.out>a}).length}
function
on_home(){if(!$('s_in').value){$('s_in').value=addDays(5);$('s_out').value=addDays(7);searchRooms()}}
function searchRooms(){
  var a=$('s_in').value,b=$('s_out').value;
  if(!a||!b||b<=a)return msg('Choose a valid arrival and departure date.',false);
  var n=days(a,b);
  $('results').innerHTML=ROOMS.map(function(r,i){
    var left=r.qty-booked(i,a,b);
    return '<div class="card"><div class="big">'+r.icon+'</div><h2>'+r.t+'</h2><p
class="muted">'+r.info+'</p><p><b>Rs. '+r.price+'</b> / night<br>'+n+' night(s): Rs.
'+(r.price*n)+'</p>'+
      '<p>'+(left>0?'<span class="ok">'+left+' room(s) available</span>':'<span class="bad">Sold
out</span>')+'</p>'+
      '<button '+(left>0?'':'disabled')+' onclick="startBooking('+i+')">Book
Now</button></div>'}).join('');
}
function startBooking(i){
  if(!me)return show('login'),msg('Please log in to book a room.',false);
  if(me.role==='admin')return msg('Log in as a guest to book.',false);
  var a=$('s_in').value,b=$('s_out').value,n=days(a,b);
  pending={room:i,in:a,out:b,guests:+$('s_g').value,total:ROOMS[i].price*n};
  show('book');$('b_sum').innerHTML='<b>'+ROOMS[i].t+'</b><br>'+a+' to '+b+' ('+n+' nights),
'+pending.guests+' guest(s)<br><b>Total: Rs. '+pending.total+'</b>';
  $('b_n').value=me.name;
}
function confirmBooking(){
  if(!$('b_n').value.trim()||$('b_c').value.replace(/\D/g,'').length<12)return msg('Enter guest name
and a valid card number.',false);

bookings.push({id:nextId++,room:pending.room,in:pending.in,out:pending.out,guests:pending.guests,name:
$('b_n').value.trim(),by:me.u,total:pending.total,status:'Confirmed'});
  show('mybookings');msg('Payment successful. Booking #'+(nextId-1)+' confirmed. A confirmation email
has been sent.');
}

// ---------------------------------------------- my bookings (3 day rule)
function canEdit(b){return b.status==='Confirmed'&&days(fmt(new Date()),b.in)>=3}
function on_mybookings(){
  var rows=bookings.filter(function(b){return b.by===me.u}).map(function(b){
    var ok=canEdit(b);
    return
'<tr><td>#'+b.id+'</td><td>'+ROOMS[b.room].t+'</td><td>'+b.in+'</td><td>'+b.out+'</td><td>Rs.
'+b.total+'</td><td class="'+(b.status==='Confirmed'?'ok':'bad')+'">'+b.status+'</td>'+
    '<td><button class="s" '+(ok?'':'disabled')+' onclick="startChange('+b.id+')">Change</button>
<button class="s red" '+(ok?'':'disabled')+'
onclick="cancelBooking('+b.id+')">Cancel</button></td></tr>'}).join('')||'<tr><td colspan="7">No
bookings yet.</td></tr>';

$('mb_table').innerHTML='<table><tr><th>ID</th><th>Room</th><th>Arrival</th><th>Departure</th><th>Tota
l</th><th>Status</th><th>Action</th></tr>'+rows+'</table>';
}
function find(id){return bookings.find(function(b){return b.id===id})}
function startChange(id){editing=find(id);show('change');$('c_sum').innerHTML='Booking #'+id+' -
'+ROOMS[editing.room].t;$('c_in').value=editing.in;$('c_out').value=editing.out}
function saveChange(){
  var a=$('c_in').value,b=$('c_out').value;
  if(!a||b<=a)return msg('Invalid dates.',false);
  if(days(fmt(new Date()),a)<3)return msg('New arrival date must be at least 3 days from
today.',false);
  if(booked(editing.room,a,b)-(editing.in<b&&editing.out>a?1:0)>=ROOMS[editing.room].qty)return
msg('No rooms available on those dates.',false);

editing.in=a;editing.out=b;editing.total=ROOMS[editing.room].price*days(a,b);show('mybookings');msg('B
ooking updated.');
```

```
}
function cancelBooking(id){if(confirm('Cancel this
booking?')){find(id).status='Cancelled';on_mybookings();msg('Booking cancelled.')}}

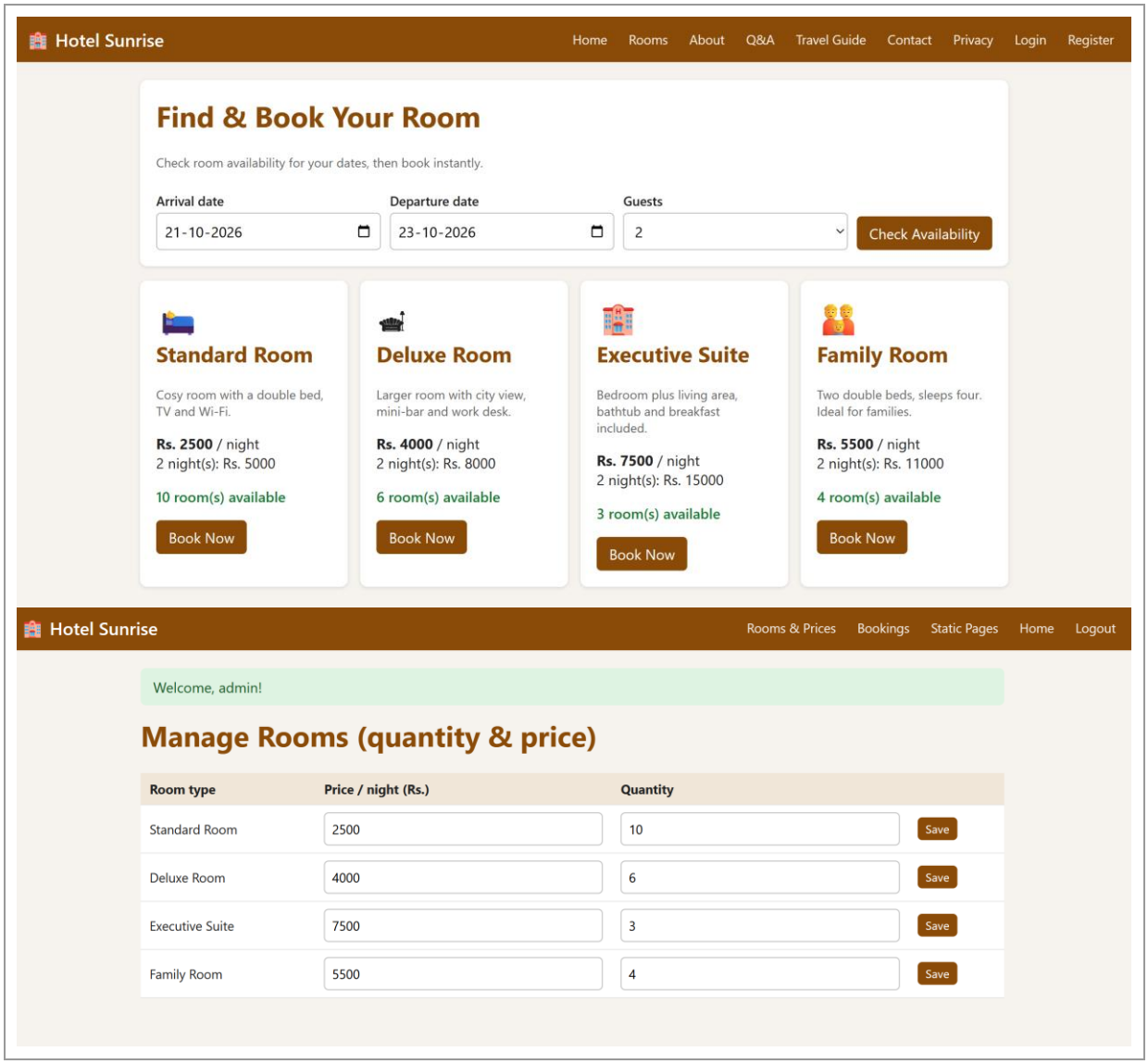
// ---------------------------------------------- admin
function on_a_rooms(){
  $('ar_table').innerHTML='<table><tr><th>Room type</th><th>Price / night
(Rs.)</th><th>Quantity</th><th></th></tr>'+ROOMS.map(function(r,i){
    return '<tr><td>'+r.t+'</td><td><input id="ap'+i+'" type="number"
value="'+r.price+'"></td><td><input id="aq'+i+'" type="number" value="'+r.qty+'"></td><td><button
class="s" onclick="saveRoom('+i+')">Save</button></td></tr>'}).join('')+'</table>';
}
function saveRoom(i){var p=+$('ap'+i).value,q=+$('aq'+i).value;if(!(p>0)||!(q>=0))return msg('Enter
valid numbers.',false);ROOMS[i].price=p;ROOMS[i].qty=q;msg(ROOMS[i].t+' updated.')}
function on_a_bookings(){

$('ab_table').innerHTML='<table><tr><th>ID</th><th>Guest</th><th>Room</th><th>Arrival</th><th>Departur
e</th><th>Total</th><th>Status</th><th>Action</th></tr>'+bookings.map(function(b){
    return
'<tr><td>#'+b.id+'</td><td>'+b.name+'</td><td>'+ROOMS[b.room].t+'</td><td>'+b.in+'</td><td>'+b.out+'</
td><td>Rs. '+b.total+'</td><td class="'+(b.status==='Confirmed'?'ok':'bad')+'">'+b.status+'</td>'+
    '<td><button class="s" onclick="adminEdit('+b.id+')">Edit</button> <button class="s red"
'+(b.status==='Cancelled'?'disabled':'')+'
onclick="adminCancel('+b.id+')">Cancel</button></td></tr>'}).join('')+'</table>';
}
function adminCancel(id){if(confirm('Cancel booking
#'+id+'?')){find(id).status='Cancelled';on_a_bookings();msg('Booking cancelled.')}}
function adminEdit(id){editing=find(id);show('a_editbook');$('e_sum').innerHTML='Booking
#'+id;$('e_n').value=editing.name;
  $('e_r').innerHTML=ROOMS.map(function(r,i){return '<option value="'+i+'"'+(i===editing.room?'
selected':'')+'>'+r.t+'</option>'}).join('');$('e_in').value=editing.in;$('e_out').value=editing.out}
function saveAdminEdit(){
  if($('e_out').value<=$('e_in').value)return msg('Invalid dates.',false);

editing.name=$('e_n').value;editing.room=+$('e_r').value;editing.in=$('e_in').value;editing.out=$('e_o
ut').value;
  editing.total=ROOMS[editing.room].price*days(editing.in,editing.out);show('a_bookings');msg('Booking
updated.');
}
var STN={rooms:'Room information',about:'About us',contactus:'Contact us',qa:'Customer service
Q&A',travel:'Local travel & shipping guide',privacy:'Privacy policy'};
function on_a_static(){$('st_sel').innerHTML=Object.keys(STN).map(function(k){return '<option
value="'+k+'">'+STN[k]+'</option>'}).join('');loadStatic()}
function loadStatic(){$('st_txt').value=STATIC[$('st_sel').value]}
function saveStatic(){STATIC[$('st_sel').value]=$('st_txt').value;msg('Page saved. Open it from the
Home menu to view.')}

// ---------------------------------------------- start
show('home');
</script>
</body>
</html>
```

7. Output



8. Result

The online hotel room booking website prototype was designed and tested successfully; all the listed pages and features work in the browser.